

Task 1.2: Detailed Review of Original C-Parameter Literature

Paper: "Infrared Safe but Infinite: Soft-Gluon Divergences Inside the Physical Region"
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1. Executive Summary

This foundational paper introduces the concept of **Sudakov shoulders** and provides the first analysis of why infrared-safe observables can produce divergent perturbative predictions at points *inside* phase space. The C-parameter in e^+e^- annihilation serves as the primary example, with its critical point at $C = 3/4$.

Key insight: The Sterman-Weinberg criteria guarantee infrared finiteness only *after* all-order resummation, not order-by-order in perturbation theory.

2. C-Parameter Definition

2.1 Linearized Momentum Tensor

The C-parameter is constructed from the linearized momentum tensor:

$$\theta^{\{\alpha\beta\}} = [\sum_i p_i^\alpha p_i^\beta / |p_i|] / [\sum_j |p_j|]$$

The eigenvalues λ_i of this 3×3 tensor satisfy:

$$0 \leq \lambda_i \leq 1, \quad \sum_i \lambda_i = 1$$

2.2 C-Parameter Formula

$$C = 3(\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1)$$

Kinematic range: $0 \leq C \leq 1$

Physical interpretation:

- $C = 0$: Perfect two-jet configuration (e.g., $\lambda_1 = 1, \lambda_2 = \lambda_3 = 0$)
- $C = 1$: Isotropic, acoplanar distribution ($\lambda_1 = \lambda_2 = \lambda_3 = 1/3$)

2.3 Three-Particle Formula

For three massless particles with energy fractions $x_i = 2p_i \cdot Q/Q^2$ (where $x_1 + x_2 + x_3 = 2$):

$$C = C_3(x_1, x_2) = 6(1 - x_1)(1 - x_2)(1 - x_3) / (x_1 x_2 x_3)$$

Critical observation: Planar events (3 particles) have one vanishing eigenvalue, restricting them to $C \leq 3/4$.

2.4 Symmetric Trijet Configuration

The maximum $C = 3/4$ for three particles occurs at:

$$x_1 = x_2 = x_3 = 2/3$$

This is the **symmetric trijet configuration** with three partons at 120° separation.

3. The Step Discontinuity at $C = 3/4$

3.1 Origin of the Step

Unlike thrust (which has an edge/kink), the C -parameter has a **step discontinuity** at $C = 3/4$ because:

1. The 3-parton phase space boundary lies at $C = 3/4$
2. The matrix element is **finite and non-zero** at this boundary
3. The phase space measure is **finite** as $C \rightarrow 3/4^-$

3.2 Leading Order Distribution

The $O(\alpha_s)$ distribution near $C = 3/4$:

$$A(C) \approx A(3/4) \theta(3/4 - C) \quad \text{as } C \rightarrow 3/4$$

where:

$$A(3/4) = (32/243) \pi \sqrt{3} M_0 = (256/243) \pi \sqrt{3} C_F \approx 5.93 C_F$$

The matrix element at the symmetric point:

$$M_0 = M(x_1 = x_2 = 2/3) = 8 C_F$$

3.3 Derivative at the Boundary

The first derivative is also finite:

$$A'(3/4) = -(8/3) A(3/4)$$

This means both $A(3/4^-)$ and $A'(3/4^-)$ are finite and non-zero, producing a true step.

4. Soft-Gluon Singularity Mechanism

4.1 General Framework

For an observable C that increases with soft-gluon emission, the $(n+1)$ -th order cross section relates to the n -th order via:

$$\hat{\sigma}^{(n+1)}(C) = \int_0^C dy \hat{\sigma}^{(n)}(C - y) [dw(z)/dz]_{z=1-y} + \dots$$

where the soft-gluon emission probability (to DL accuracy) is:

$$dw(z)/dz = a [(1/(1-z)) \ln(1/(1-z))] +$$

with $a = K \alpha_s(Q)/\pi$.

4.2 Singularity at the Step

When $\hat{\sigma}^{(n)}$ has a step at $C = C_0$, the next order develops a **double-logarithmic divergence**:

$$\begin{aligned} \hat{\sigma}^{(n+1)}_+(C) - \hat{\sigma}^{(n+1)}_-(C_0) = & -(a/2) \ln^2(C - C_0) \times \{ [\hat{\sigma}^{(n)}_+(C_0) - \hat{\sigma}^{(n)}_-(C_0)] \\ & + (C - C_0) [\hat{\sigma}^{(n)'}_+(C_0) - \hat{\sigma}^{(n)'}_-(C_0)] + \dots \} + \dots \end{aligned}$$

4.3 Contrast with Edge (Thrust)

For thrust at $\tau = 1/3$:

- Distribution is **continuous** but derivative is **discontinuous**
- This produces a **cusplike** at next order (divergent derivative, finite distribution)
- The edge becomes more step-like at each higher order

For C-parameter at $C = 3/4$:

- Distribution itself is **discontinuous** (step)
- This produces **divergent distribution** at next order
- More severe singularity structure

5. NLO Results for C-Parameter

5.1 Perturbative Expansion

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$$(1/\sigma_0)(d\sigma/dC) = \bar{\alpha}_s A(C) + \bar{\alpha}_s^2 B(C) + O(\alpha_s^3)$$

\\

where $\bar{\alpha}_s = \alpha_s(Q)/2\pi$.

5.2 Below the Shoulder ($C < 3/4$)

$B_-(C)$ is smooth with:

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$$B_-(3/4) = 230 \pm 10 \quad (\text{from EVENT Monte Carlo})$$

\\

5.3 Above the Shoulder ($C > 3/4$)

The NLO coefficient diverges logarithmically:

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$$B_+(C) = A(3/4) [(2C_F + C_A)(1 - \delta) \ln^2 \delta + (3C_F + (11/6)C_A - (1/3)N_f) \ln \delta] + h(C)$$

\\

where:

\\

$$\delta = (8/3)(C - 3/4)$$

\\

and:

\\

$$h(3/4) = 146 \pm 3 \quad (\text{from EVENT Monte Carlo})$$

\\

5.4 Color Structure

****Double-log coefficient:**** $(2C_F + C_A) = 2C_F + C_A$

- This matches the general prediction from Eq. (2.9) with $a = (2C_F + C_A)\alpha_s(Q)/\pi$

****Single-log coefficient:**** $(3C_F + (11/6)C_A - (1/3)N_f)$

- Includes collinear (non-soft) contributions

6. Double-Logarithmic Resummation

6.1 Sudakov Form Factor at Exclusive Boundary

Near $C = 0$ (two-jet limit):

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$$\hat{\sigma}^{(\infty)}(C) = \exp\{-(a/2) \ln^2 C\} \hat{\sigma}^{(n_0)}(C)$$

This is the standard Sudakov suppression.

6.2 Sudakov Shoulder at Critical Point

Near $C = C_0 = 3/4$:

$$\hat{\sigma}^{(\infty)}_+(C) - \hat{\sigma}^{(\infty)}_-(C_0) = \exp\{-(a/2) \ln^2(C - C_0)\} [\hat{\sigma}^{(n_0)}_+(C_0) - \hat{\sigma}^{(n_0)}_-(C_0)]$$

****Key result:**** Rather than suppressing the divergence, Sudakov resummation ****suppresses the step itself****, producing a smooth, infinitely differentiable distribution.

6.3 Resummed C-Parameter Distribution

At DL accuracy, approximating $A(C) \approx A(3/4)\theta(3/4 - C)$:

$$(1/\sigma)(d\sigma/dC) \approx \bar{\alpha}_s A(3/4) \{\theta(3/4 - C) + \theta(C - 3/4)(1 - \exp[-2A^{(1)} \bar{\alpha}_s \ln^2(C - 3/4)])\}$$

where:

$$A^{(1)} = C_F + (1/2)C_A$$

6.4 Properties of the Shoulder

1. ****Finite at $C = 3/4$:** The distribution is continuous
2. ****Smooth (C^∞):** All derivatives exist and are finite
3. ****Steepness:** Decreases with increasing $A^{(1)}$ or α_s
4. ****QCD vs Abelian:** The shoulder is broader in QCD due to gluon fragmentation ($A^{(1)} \sim 2C_F$ in QCD vs C_F in QED)

7. Comparison: Step vs Edge Resummation

7.1 Step (C-parameter at $C = 3/4$)

****Lower order:**** $\hat{\sigma}^{(n)}(C)$ has discontinuity at C_0
****Higher order:**** $\hat{\sigma}^{(n+1)}(C)$ diverges as $\ln^2(C - C_0)$ for $C > C_0$
****After resummation:**** Step is smoothed into continuous shoulder

7.2 Edge (Thrust at $\tau = 1/3$)

****Lower order:**** $\hat{\sigma}^{(n)}(C)$ continuous, $\hat{\sigma}^{(n)'}(C)$ discontinuous
****Higher order:**** $\hat{\sigma}^{(n+1)'}(C)$ diverges as $\ln^2(C - C_0)$
****After resummation:**** Apply DL resummation to derivative:

$$\hat{\sigma}^{(\infty)}_+(C) = \hat{\sigma}^{(\infty)}_-(C_0) + (C - C_0)\hat{\sigma}^{(\infty)'}_-(C_0) + [\hat{\sigma}^{(n_0)'}_+(C_0) - \hat{\sigma}^{(n_0)'}_-(C_0)] \int_{C_0}^C dx e^{-(a/2)\ln^2 x}$$

8. Key Formulas Summary

Quantity	Formula	Value
C (3 particles)	$6(1-x_1)(1-x_2)(1-x_3)/(x_1x_2x_3)$	-
$C_{\max}$ (3 particles)	$3/4$	0.75
Symmetric point	$x_1 = x_2 = x_3 = 2/3$	-
$A(3/4)$	$(256/243)\pi\sqrt{3} C_F$	$\approx 5.93 C_F$
$A'(3/4)$	$-(8/3) A(3/4)$	$\approx -15.8 C_F$

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| DL coefficient |  $A^{\{1\}} = C_F + C_A/2 \approx 2.83$  (QCD) |
| NLO  $\delta$  variable |  $(8/3)(C - 3/4)$  | - |
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9. Implications for NLL Resummation Project

9.1 What This Paper Establishes

1. **The critical point is $C = 3/4$** (symmetric trijet)
2. **The discontinuity is a step** (not an edge like thrust)
3. **DL resummation works** and produces a Sudakov shoulder
4. **Color structure:** The coefficient is $(2C_F + C_A)$ for the double logs

9.2 What Remains to be Done (for NLL)

1. **Single-logarithmic terms:** Only DL resummation is performed here
2. **Factorization theorem:** Not derived in SCET framework
3. **Soft function:** Not computed; 6-sextant decomposition not discussed
4. **Left shoulder:** This paper focuses on $C > 3/4$; need to determine if $C < 3/4$ also requires resummation
5. **Matching:** No matching to fixed order discussed
6. **Running coupling:** Not included in DL formula

9.3 Differences from BSZ (2205.05702) Approach

Aspect	Catani-Webber	BSZ
Framework	Full QCD + DL resummation	SCET factorization + NLL
Accuracy	Double-log only	Next-to-leading log
Soft function	Not computed	6-sextant decomposition
Factorization	Heuristic convolution	Rigorous factorization theorem
Power corrections	Not discussed	Analyzed in detail
Landau pole	Not discussed	Identified and analyzed

10. Connection to Invariant Formulation

10.1 C in Terms of Invariants

For 3 massless partons with $s_{ij} = (p_i + p_j)^2/Q^2$ and $s_{12} + s_{13} + s_{23} = 1$:

The relationship between C and the invariants is:

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$$C = 6 s_{12} s_{13} s_{23}$$

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Verification: At symmetric point $s_{12} = s_{13} = s_{23} = 1/3$:

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$$C = 6 \times (1/3)^3 \times 3 = 6/27 \times 3 = 6/9 = \dots$$

```

Wait, let me recalculate. With the energy fractions:

$$- x_1 = 1 - s_{23}, x_2 = 1 - s_{13}, x_3 = 1 - s_{12}$$

$$- \text{At symmetric point: } x_i = 2/3 \rightarrow s_{ij} = 1/3$$

Then:

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$$C = 6(1-x_1)(1-x_2)(1-x_3)/(x_1x_2x_3) = 6(1/3)^3/(2/3)^3 = 6 \times (1/27)/(8/27) = 6/8 = 3/4 \checkmark$$

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10.2 Task 1.7 Preview

The relation $C = 6 s_{12} s_{13} s_{23}$ should be verified in Task 1.7. Note that this differs from

the BSZ paper's statement that thrust $\tau = \min(s_{12}, s_{13}, s_{23})$.

11. Physical Picture

11.1 Why the Step Exists

1. Three massless partons can only reach $C = 3/4$ (planar constraint)
2. At $C = 3/4$, the matrix element $M_0 = 8C_F$ is finite
3. The phase space doesn't vanish (unlike thrust at $\tau = 1/3$)
4. Result: The distribution has a finite, non-zero value at the boundary

11.2 Why the Divergence Appears at NLO

1. Virtual corrections are confined to $C < 3/4$ (3-parton kinematics)
2. Real emission of 4th parton can reach $C > 3/4$
3. At $C = 3/4^+$, real emission is present but virtual corrections are "missing"
4. The imbalance produces logarithmic divergence

11.3 Why Resummation Fixes It

1. Multiple soft emissions can push C above or below $3/4$
2. The probability of remaining exactly at $C = 3/4$ vanishes
3. The Sudakov factor $\exp(-a \ln^2 \delta)$ suppresses the step
4. Result: Smooth shoulder structure

12. Notation Mapping

Catani-Webber	BSZ	Meaning
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C	C	C -parameter
$C_0 = 3/4$	$C_{sh} = 3/4$	Critical point
$\delta = (8/3)(C - 3/4)$	$\Delta = C - 3/4$	Right shoulder variable
$-c = 3/4 - C$		Left shoulder variable
$A^{\{1\}} = C_F + C_A/2$	$(2C_F + C_A)/2$	DL anomalous dimension
$\bar{\alpha}_s = \alpha_s/2\pi$	$\alpha_s/4\pi$	Coupling convention

References

1. **C-parameter original:** Parisi (1978), Donoghue-Low-Pi (1979), Ellis-Ross-Terrano (1981)
2. **Thrust resummation:** Catani-Turnock-Webber-Trentadue, NPB 407 (1993) 3
3. **EVENT Monte Carlo:** Kunszt-Nason-Marchesini-Webber, CERN 89-08
4. **Sequel paper:** Catani-Webber, Cavendish-HEP-97/16 (C-parameter resummation)

Document prepared for C-parameter Sudakov shoulder research project
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